

FLOWMATE CA

Purchase Order Creation API Integration Guide

Technical contract for external systems creating PO workflow requests

Version: 1.0

Prepared: 24 September 2026

Target service: FlowmateCAService (OData V4)

Audience: Solution architects, integration developers, API testers and support teams

Document basis

Generated from the checked-in OpenAPI 3.0.2 specification and reconciled with the current Flowmate CA CDS model, service implementation and UI request contract.

1. Purpose and scope

This guide describes the complete application-level contract for an external system to create a Purchase Order workflow request in Flowmate CA. It covers authentication, prerequisite value-help APIs, request creation, line items, file attachment upload, retrieval, status tracking and optional workflow operations.

Scope boundary

The API creates a **Flowmate CA workflow request** of type **PURCHASE_ORDER**. It does not directly create an SAP S/4HANA purchase order. The current **sendToS4** action is an integration boundary and returns a readiness message until the customer destination and mapping are enabled.

Contract hierarchy

Priority	Source	Use
1	Current CDS model and service implementation	Runtime entity fields, action parameters, validation and processing behavior
2	Current UI form definitions	Fields marked mandatory for interactive PO creation and attachment categories
3	Combined Swagger / OpenAPI 3.0.2	Published paths, standard OData request and response shapes

Known Swagger drift

The checked-in Swagger omits **processorTeamCode** from NewRequestInput and omits newer PO header/item fields. This guide includes those current runtime fields and labels them accordingly. The integration team should not generate its final client solely from the current Swagger until it is regenerated from the latest Flowmate CA model.

Environment variables

Variable	Description	Example / source
TOKEN_URL	XSUAA OAuth token endpoint	https://arize-qas-9339ozek.authentication.ap11.hana.ondemand.com/oauth/token
CLIENT_ID	flowmate-api-auth service key client ID	Intentionally excluded - obtain securely from the service key
CLIENT_SECRET	flowmate-api-auth service key secret	Intentionally excluded - obtain securely from the service key
CA_BASE_URL	Direct Flowmate CA service application route	https://dialog-axiata-plc-arize-qas-9339ozek-org-a-qas-org-b-te6cc03f10.cfa.pps.ap11.hana.ondemand.com
SERVICE_ROOT	OData V4 root	\${CA_BASE_URL}/odata/v4/flowmate-ca
MASTER_ROOT	CA master-data root	\${CA_BASE_URL}/odata/v4/flowmate-ca-master

QAS endpoint directory

These URLs were resolved from the Dialog Axiata QAS Cloud Foundry space on 24 September 2026. Application routes can change after route recreation, blue-green deployment or environment migration; verify them before production onboarding.

Endpoint	QAS URL	Use
OAuth token	https://arize-qas-9339ozek.authentication.ap11.hana.ondemand.com/oauth/token	POST client_credentials token request
Flowmate CA direct API	https://dialog-axiata-plc-arize-qas-9339ozek-org-a-qas-org-b-te6cc03f10.cfapps.ap11.hana.ondemand.com	Recommended host for system integration
PO service root	https://dialog-axiata-plc-arize-qas-9339ozek-org-a-qas-org-b-te6cc03f10.cfapps.ap11.hana.ondemand.com/odata/v4/flowmate-ca	All PO request and attachment APIs
CA master-data root	https://dialog-axiata-plc-arize-qas-9339ozek-org-a-qas-org-b-te6cc03f10.cfapps.ap11.hana.ondemand.com/odata/v4/flowmate-ca-master	Shared value-help endpoints exposed by CA
OData metadata	https://dialog-axiata-plc-arize-qas-9339ozek-org-a-qas-org-b-te6cc03f10.cfapps.ap11.hana.ondemand.com/odata/v4/flowmate-ca/\$metadata	Machine-readable deployed OData contract
Identity check	https://dialog-axiata-plc-arize-qas-9339ozek-org-a-qas-org-b-te6cc03f10.cfapps.ap11.hana.ondemand.com/odata/v4/flowmate-ca/getCurrentUser()	Verify token and user provisioning
Standalone CA UI	https://dialog-axiata-plc-arize-qas-9339ozek-org-a-qas-org-b-te229cbf03.cfapps.ap11.hana.ondemand.com	Browser UI through standalone approuter
Combined Swagger UI	https://dialog-axiata-plc-arize-qas-9339ozek-org-a-qas-org-b-te48c9fbef.cfapps.ap11.hana.ondemand.com/api-docs	Interactive combined API reference on Flowmate service

Which host should the integration team call?

Use the **Flowmate CA direct API** host with the bearer token for system-to-system calls. The standalone CA UI route is intended for interactive browser sessions and SSO.

2. End-to-end integration sequence

Step	Operation	Outcome
1	Authenticate	Obtain an XSUAA bearer token with client credentials.
2	Verify identity	Call getCurrentUser. The technical principal must resolve to an active shared-master-data user.
3	Resolve codes	Read the active request variant, priority, processor team and all PO master-data values.
4	Create request	POST createRequest with input.details encoded as one JSON string.
5	Upload files	For every file, create attachment metadata, then PUT the binary stream to /content.
6	Read back	GET the request with \$expand=purchaseOrder(\$expand=items),attachments,stepInstances,tasks.
7	Track status	Poll the request/tasks or consume an agreed notification mechanism.
8	Optional handoff	Call sendToS4 only after the S/4 destination and mapping have been enabled.

Recommended sequence diagram

```

sequenceDiagram
    participant External system
    participant XSUAA
    participant Flowmate CA
    participant Common master DB

    External system->>XSUAA: -- client token -->
    XSUAA-->>External system: <-- access_token --
    XSUAA->>Flowmate CA: -- getCurrentUser ----->
    Flowmate CA-->>XSUAA: <-- user lookup ----->
    XSUAA->>Common master DB: <-- current user / 403 ----->
    Common master DB-->>XSUAA: <-- value-help GETs ----->
    XSUAA->>Flowmate CA: <-- active codes and UUIDs ----->
    Flowmate CA-->>XSUAA: <-- POST createRequest ----->
    XSUAA->>Common master DB: <-- request/details -->
    Common master DB-->>XSUAA: <-- request ID + referenceNumber ----->
    XSUAA->>Flowmate CA: <-- POST Attachments ----->
    Flowmate CA-->>XSUAA: <-- attachment ID ----->
    XSUAA->>Common master DB: <-- PUT Attachments(ID)/content ----->
    Common master DB-->>XSUAA: <-- GET Requests(ID)?$expand=... ----->
    XSUAA->>Flowmate CA: <-- workflow state ----->
  
```

Transaction behavior

Request header, PO details, PO items, initial workflow steps and history are created in the same CAP transaction. Attachment binaries are uploaded afterward and are therefore separate operations.

3. Authentication and authorization

Production requests use an XSUAA JWT bearer token. The intended system-to-system credential is the service key of **flowmate-api-auth** (xsappname **flowmate-api-client**). Never distribute the raw service-key JSON in email or source control.

3.1 Obtain an access token

```
curl --request POST "$TOKEN_URL" \
  --user "$CLIENT_ID:$CLIENT_SECRET" \
  --header "Content-Type: application/x-www-form-urlencoded" \
  --data "grant_type=client_credentials"
```

Representative success response:

```
{
  "access_token": "eyJ...redacted...",
  "token_type": "bearer",
  "expires_in": 43199,
  "scope": "flowmate-ca.CAAdmin"
}
```

3.2 Standard headers

Header	Value	When
Authorization	Bearer <access_token>	Every protected request
Accept	application/json	JSON APIs
Content-Type	application/json	JSON POST/PATCH
Content-Type	Actual MIME type, e.g. application/pdf	Attachment content PUT
X-CSRF-Token	Token returned by a Fetch request	If calling through a route that enforces CSRF

3.3 Provisioning prerequisite

Potential 403 condition

The create handler calls **_ensureCurrentUser**. It lowercases the JWT login/email and matches it against an active record in the shared Users table. A client-credentials principal that is not represented there receives HTTP 403: *Your user is not provisioned in the shared Flowmate master data service*. Confirm the technical principal mapping before integration testing.

Identity verification

```
curl "$SERVICE_ROOT/getCurrentUser()" \
  --header "Authorization: Bearer $ACCESS_TOKEN" \
  --header "Accept: application/json"

{
  "ID": "11111111-1111-4111-8111-111111111111",
  "displayName": "PO Integration Client",
  "email": "po.integration@example.com",
  "isAdmin": true
}
```

4. API endpoint catalogue

The following endpoints are the recommended integration surface for PO request creation. Standard OData query options include \$select, \$filter, \$orderby, \$top, \$skip, \$count and \$expand where exposed by Swagger.

Method	Path	Purpose
POST	{TOKEN_URL}	Obtain bearer token
GET	/odata/v4/flowmate-ca/\$metadata	Read OData metadata
GET	/odata/v4/flowmate-ca/getCurrentUser()	Verify principal provisioning
GET	/odata/v4/flowmate-ca/RequestTypes	Resolve PURCHASE_ORDER
GET	/odata/v4/flowmate-ca/RequestVariants	Resolve PO_CAPEX_IM / PO_CAPEX_FM / PO_OPEX
GET	/odata/v4/flowmate-ca/Priorities	Resolve priority code
GET	/odata/v4/flowmate-ca/Teams	Resolve processor team code
GET	/odata/v4/flowmate-ca/Vendors	Resolve vendor UUID and code
GET	/odata/v4/flowmate-ca/PurchaseOrderTypes	Resolve PO type code
GET	/odata/v4/flowmate-ca/ProcurementCategories	Resolve category code
GET	/odata/v4/flowmate-ca/PurchasingOrganizations	Resolve purchasing organization
GET	/odata/v4/flowmate-ca/Currencies	Resolve currency
GET	/odata/v4/flowmate-ca/PaymentTerms	Resolve payment terms
GET	/odata/v4/flowmate-ca/UnitsOfMeasure	Resolve item UoM
GET	/odata/v4/flowmate-ca-master/Divisions	Resolve division
GET	/odata/v4/flowmate-ca-master/Users	Resolve delegated/divisional user UUID
GET	/odata/v4/flowmate-ca-master/Sites	Resolve site/campaign location
GET	/odata/v4/flowmate-ca-master/DocumentTypes	Resolve document type
GET	/odata/v4/flowmate-ca-master/PurchasingGroups	Resolve purchasing group
GET	/odata/v4/flowmate-ca-master/Plant	Resolve plant
GET	/odata/v4/flowmate-ca-master/Wbs	Resolve WBS element
POST	/odata/v4/flowmate-ca/createRequest	Create request, details and items
POST	/odata/v4/flowmate-ca/Attachments	Create attachment metadata
PUT	/odata/v4/flowmate-ca/Attachments({ID})/content	Upload attachment bytes
GET	/odata/v4/flowmate-ca/Requests({ID})	Read request and expanded workflow state
GET	/odata/v4/flowmate-ca/PurchaseOrderDetails({ID})	Read PO details
GET	/odata/v4/flowmate-ca/PurchaseOrderItems	Read/filter PO line items
POST	/odata/v4/flowmate-ca/submitRequest	Legacy explicit submit action; not used by current UI
POST	/odata/v4/flowmate-ca/sendToS4	Optional S/4 integration boundary

5. Master-data and code resolution

Do not hard-code UUIDs. Cache stable codes only, refresh active value helps periodically, and reject a transaction locally when a referenced value is inactive or absent.

Representative lookup requests

```
GET $SERVICE_ROOT/RequestVariants?$select=code,name&$filter=requestType_code eq 'PURCHASE_ORDER' and isActive eq true&$orderby=sortOrder
GET $SERVICE_ROOT/Priorities?$select=code,name&$filter=isActive eq true&$orderby=sortOrder
GET $SERVICE_ROOT/Teams?$select=ID,teamCode,name&$filter=isActive eq true
GET $SERVICE_ROOT/Vendors?$select=ID,vendorCode,vendorName&$filter=isActive eq true
GET $SERVICE_ROOT/PurchaseOrderTypes?$select=code,name&$filter=isActive eq true
GET $MASTER_ROOT/Plant?$select=ID,plantCode,plantName&$filter=isActive eq true
```

Supported PO variants

Code	Name	Current workflow
PO_CAPEX_IM	Capex IM	Request details -> S/4 data validation -> PO creation
PO_CAPEX_FM	Capex FM	Request details -> budget confirmation -> S/4 validation -> PO creation
PO_OPEX	Opex	Request details -> budget confirmation -> S/4 validation -> PO creation

OData collection response shape

```
{
  "@odata.context": "$metadata#Vendors",
  "value": [
    {
      "ID": "aaaaaaaa-aaaa-4aaa-8aaa-aaaaaaaaaaaa",
      "vendorCode": "V0004711",
      "vendorName": "Example Supplies Ltd",
      "isActive": true
    }
  ]
}
```

Value conventions

Association foreign keys use CAP-generated names such as **vendor_ID**, **currency_code** and **unitOfMeasure_code**. Codes are case-sensitive at the database query level unless the service explicitly normalizes them.

6. Create-request API

POST /odata/v4/flowmate-ca/createRequest

The request body contains one property named **input**. The nested **details** property is not a JSON object in the action contract; it must be a JSON-encoded string. The service parses that string and writes PurchaseOrderDetails and PurchaseOrderItems.

Field	CDS type	Required	Example	Rules
requestTypeCode	String(40)	Yes	PURCHASE_ORDER	Must identify an active RequestType
requestVariantCode	String(40)	Yes*	PO_OPEX	Current UI requires it; service validates ownership when supplied
title	String(255)	Yes*	Opex PO - network maintenance	Current UI and practical integration requirement
description	LargeString	No	Quarterly maintenance procurement	Free text
priorityCode	String(40)	Yes*	HIGH	Current UI requires it; service defaults MEDIUM if omitted
dueDate	Date	No	2026-10-15	ISO 8601 YYYY-MM-DD
predecessorId	UUID	No	null	Existing predecessor request UUID
processorTeamCode	String(40)	Yes*	PROCUREMENT	Present in current CDS/UI; missing from checked-in Swagger
details	LargeString	Yes	{"selectionMode":...}	JSON serialized as a string

Required legend

Yes means enforced by the model or handler. **Yes*** means required by the current Flowmate CA creation UI/integration contract but not consistently enforced by backend validation. External callers must validate these fields before calling.

7. Purchase-order header field specification

Field	Type	Requirement	Dummy value	Meaning / validation
selectionMode	String(20)	Required*	SINGLE MULTIPLE	Selection mode
purchaseOrderType_code	String(40)	Required*	ZNB	PurchaseOrderTypes.code
contractBased	Boolean	Optional	false	Contract-based indicator
contractNumber	String(80)	Optional	CTR-2026-0042	Contract reference
procurementCategory_code	String(40)	Required*	IT_SERVICES	ProcurementCategories.code
division	String(80)	Optional	TECH	Division code
delegatedUser_ID	UUID	Optional	2222...	Shared Users.ID
specialPersonArea	String(80)	Optional	IT	FTK, ENGINEERING, IT, DBS, MARKETING, FOC or DNS
arNumber	String(80)	Optional	AR-009876	AR reference
wbsElement	String(80)	Optional	WBS-100023	WBS code
documentType	String(40)	Optional	PO	Document type code
taxApplicable	Boolean	Optional	true	Tax indicator
clearanceChargeApplicable	Boolean	Optional	false	Clearance charge indicator
materialImported	Boolean	Optional	false	Imported material indicator
sesRequired	Boolean	Optional	true	Create SES successor flag
paymentRequestRequired	Boolean	Optional	false	Create payment successor flag
poHeaderText	LargeString	Optional	Network maintenance PO	Header text
campaignLocationCode	String(100)	Optional	SITE-001	Site/location code
procurementDescription	LargeString	Required*	Managed network maintenance	Procurement description
vendor_ID	UUID	Optional	aaaa...	Shared Vendors.ID
vendorCode	String(40)	Required*	V0004711	Vendor business code; current model field
companyCode	String(40)	Optional	1000	Company code
purchasingGroup	String(40)	Optional	P01	Purchasing group code
purchasingOrganization_code	String(40)	Optional	ORG01	PurchasingOrganizations.code
siteld	String(80)	Optional	SITE-001	Site identifier
currency_code	String(40)	Required*	LKR	Currencies.code
plant_code	String(40)	Required*	PL01	Shared Plant.plantCode
totalValue	Decimal(17,2)	Required*	1250000.00	PO value excluding tax
poValueWithTaxes	Decimal(17,2)	Optional	1475000.00	Current model; absent from checked-in Swagger
procurementValue	Decimal(17,2)	Optional	1250000.00	Current model; absent from checked-in Swagger
paymentTerms_code	String(40)	Optional	NET30	PaymentTerms.code
remarks	LargeString	Optional	Deliver in October	Remarks
termsOfDelivery	LargeString	Required*	DAP Colombo	Delivery terms
warranty	LargeString	Required*	12 months	Warranty terms
paymentMilestone	String(120)	Optional	100% after acceptance	Payment milestone
approvedPoRecipientEmail	String(255)	Required*	po.receiver@example.com	Approved PO recipient
invoiceBoqAttachment	String(255)	Required*	quotation.pdf	Filename; binary uploaded separately
emailApprovalAttachment	String(255)	Required*	approval-email.pdf	Filename; binary uploaded separately
summary	LargeString	Optional	Opex maintenance PO	Summary
psrRefNo	String(80)	Required*	PSR-2026-011	PSR reference
systemContractId	String(80)	Required*	OA-550001	Contract/agreement ID
divisionalUser_ID	UUID	Required*	3333...	Shared Users.ID
procurementApprovalAttachment	String(255)	Required*	procurement-approval.pdf	Filename; binary uploaded separately
supportiveDocumentAttachment	String(255)	Required*	supporting-document.pdf	Filename; binary uploaded separately

8. Purchase-order line-item specification

Pass line items in **details.items**. The service creates a PurchaseOrderItems row for each array entry and assigns sequential itemNo values beginning with 1.

Field	Type	Dummy value	Meaning / rule
itemNo	Integer	1	Generated by createRequest; caller value is overwritten
itemType	String(30)	MATERIAL	Optional runtime field
materialOrService	String(80)	MAT-100045	Material or service code
description	String(180)	Router maintenance kit	Line description
quantity	Decimal(15,3)	10.000	Quantity
unitOfMeasure_code	String(40)	EA	UnitsOfMeasure.code; supported by model
unitPrice	Decimal(17,2)	125000.00	Unit price
taxCode	String(40)	VAT18	Tax code
deliveryDate	Date	2026-10-15	YYYY-MM-DD
wbsElement	String(80)	WBS-100023	WBS/cost center/budget code
currency	String(40)	LKR	Line currency
contractNo	String(80)	CTR-2026-0042	Contract number
campaignLocationCode	String(100)	SITE-001	Campaign/location
siteId	String(80)	SITE-001	Site
plant	String(40)	PL01	Plant code
itemCategory	String(40)	STANDARD	Item category
accountAssignment	String(80)	K	Account assignment
materialGroup	String(40)	MG-IT	Material group

UI versus model

The UI currently exposes 13 item columns. The database model also supports itemType, description, unitOfMeasure_code and deliveryDate. None of the PO item columns is currently marked mandatory in the UI; the integrating system should enforce its own business minimum.

9. Complete dummy create payload

The inner object is shown compactly because it is transmitted as an escaped string in **input.details**. Programmatic clients should call `JSON.stringify/details` serialization rather than manually escaping it.

```
{
  "input": {
    "requestTypeCode": "PURCHASE_ORDER",
    "requestVariantCode": "PO_OPEX",
    "title": "Opex PO - network maintenance",
    "description": "Quarterly network maintenance procurement",
    "priorityCode": "HIGH",
    "dueDate": "2026-10-15",
    "predecessorId": null,
    "processorTeamCode": "PROCUREMENT",
    "details": "{ \"selectionMode\": \"SINGLE\", \"purchaseOrderType_code\": \"ZNB\", \"contractBased\": false, \"contractNumber\": \"CTR-2026-0042\", \"procurementCategory_code\": \"IT_SERVICES\", \"division\": \"TECH\", \"specialPersonArea\": \"IT\", \"arNumber\": \"AR-009876\", \"wbsElement\": \"WBS-100023\", \"documentType\": \"PO\", \"taxApplicable\": true, \"clearanceChargeApplicable\": false, \"materialImported\": false, \"sesRequired\": true, \"paymentRequestRequired\": false, \"poHeaderText\": \"Network maintenance PO\", \"campaignLocationCode\": \"SITE-001\", \"procurementDescription\": \"Managed network maintenance services for Q4\", \"vendor_ID\": \"aaaaaaaa-aaaa-4aaa-8aaa-aaaaaaaaaaaa\", \"vendorCode\": \"V0004711\", \"companyCode\": \"1000\", \"purchasingGroup\": \"P01\", \"purchasingOrganization_code\": \"ORG01\", \"siteId\": \"SITE-001\", \"currency_code\": \"LKR\", \"plant_code\": \"PL01\", \"totalValue\": 1250000.0, \"poValueWithTaxes\": 1475000.0, \"procurementValue\": 1250000.0, \"paymentTerms_code\": \"NET30\", \"termsOfDelivery\": \"DAP Colombo\", \"warranty\": \"12 months\", \"paymentMilestone\": \"100% after acceptance\", \"approvedPoRecipientEmail\": \"po.receiver@example.com\", \"invoiceBoqAttachment\": \"quotation.pdf\", \"emailApprovalAttachment\": \"approval-email.pdf\", \"summary\": \"Opex maintenance PO\", \"psrRefNo\": \"PSR-2026-011\", \"systemContractId\": \"OA-550001\", \"divisionalUser_ID\": \"33333333-3333-4333-8333-333333333333\", \"procurementApprovalAttachment\": \"procurement-approval.pdf\", \"supportiveDocumentAttachment\": \"supporting-document.pdf\", \"items\": [{ \"materialOrService\": \"MAT-100045\", \"description\": \"Router maintenance kit\", \"quantity\": 10, \"unitOfMeasure_code\": \"EA\", \"unitPrice\": 125000, \"taxCode\": \"VAT18\", \"deliveryDate\": \"2026-10-15\", \"wbsElement\": \"WBS-100023\", \"currency\": \"LKR\", \"contractNo\": \"CTR-2026-0042\", \"campaignLocationCode\": \"SITE-001\", \"siteId\": \"SITE-001\", \"plant\": \"PL01\", \"itemCategory\": \"STANDARD\", \"accountAssignment\": \"K\", \"materialGroup\": \"MG-IT\" } ] }"
  }
}
```

cURL example

```
curl --request POST "$SERVICE_ROOT/createRequest" \
  --header "Authorization: Bearer $ACCESS_TOKEN" \
  --header "Accept: application/json" \
  --header "Content-Type: application/json" \
  --data @create-po-request.json
```

Representative create response

Persist both **ID** and **referenceNumber**. The UUID is required for navigation and attachment calls; the reference number is the business-facing identifier.

```
{
  "ID": "44444444-4444-4444-8444-444444444444",
  "referenceNumber": "CA-PURC-260924-381204",
  "requestType_code": "PURCHASE_ORDER",
  "requestVariant_code": "PO_OPEX",
  "title": "Opex PO - network maintenance",
  "requesterName": "PO Integration Client",
  "requesterEmail": "po.integration@example.com",
  "ownerTeamName": "Procurement",
  "status_code": "DRAFT",
  "priority_code": "HIGH",
  "currentStep": 0,
  "dueDate": "2026-10-15"
}
```

10. Attachment upload APIs

Files are not embedded in createRequest. For each file, execute two calls after a successful request creation.

Field	Type	Required	Notes
ID	UUID	No	Generated if omitted
request_ID	UUID	Yes	ID returned by createRequest
filename	String	Yes*	Original filename including extension
category	String(60)	Yes*	Logical attachment category
note	String	No	Optional note
mimeType	String	No	Derived from filename; do not send in metadata
content	Binary stream	Second call	Uploaded to /content, not JSON metadata

Recommended categories for PO creation

Details field	Attachment category
invoiceBoqAttachment	INVOICE_BOQ_ATTACHMENT
emailApprovalAttachment	EMAIL_APPROVAL_ATTACHMENT
procurementApprovalAttachment	PROCUREMENT_APPROVAL_ATTACHMENT
supportiveDocumentAttachment	SUPPORTIVE_DOCUMENT_ATTACHMENT
General supporting files	SUPPORTING_DOCUMENT

Step A - create metadata

```
curl --request POST "$SERVICE_ROOT/Attachments" \
  --header "Authorization: Bearer $ACCESS_TOKEN" \
  --header "Content-Type: application/json" \
  --data '{
    "request_ID": "44444444-4444-4444-8444-444444444444",
    "filename": "quotation.pdf",
    "category": "INVOICE_BOQ_ATTACHMENT"
  }'
```

Step B - upload binary content

```
curl --request PUT "$SERVICE_ROOT/Attachments(55555555-5555-4555-8555-555555555555)/content" \
  --header "Authorization: Bearer $ACCESS_TOKEN" \
  --header "Content-Type: application/pdf" \
  --data-binary @quotation.pdf
```

Atomicity warning

If attachment upload fails, the PO workflow request still exists. The caller must retry the failed file, record a compensating error, or delete the draft through an approved operational process.

11. Read-back and tracking APIs

Read the created aggregate rather than assuming every submitted value was accepted. Use \$select for polling and \$expand for diagnostic or reconciliation reads.

Full reconciliation read

```
GET $SERVICE_ROOT/Requests(44444444-4444-4444-8444-444444444444)?
$select=ID,referenceNumber,status_code,currentStep,externalObjectId,externalStatus&
$expand=purchaseOrder($expand=items),attachments,stepInstances,tasks,history
```

Lightweight status poll

```
GET $SERVICE_ROOT/Requests(44444444-4444-4444-8444-444444444444)?
$select=ID,referenceNumber,status_code,currentStep,modifiedAt,externalObjectId,externalStatus
```

Useful navigation paths

Path	Purpose
/Requests({ID})/purchaseOrder	PO header details
/Requests({ID})/purchaseOrder/items	PO line items through navigation
/Requests({ID})/attachments	Uploaded file metadata
/Requests({ID})/stepInstances	Guided process progress
/Requests({ID})/tasks	Operational tasks
/Requests({ID})/history	Audit history
/PurchaseOrderItems?\$filter=details_ID eq {detailsId}	Direct item collection query

Status fields

Field	Values / meaning
status_code	DRAFT, SUBMITTED, IN_PROGRESS, SENT_BACK, COMPLETED
currentStep	Current workflow step number
externalObjectId	External/SAP object identifier when populated
externalStatus	External/SAP integration state when populated
submittedAt / completedAt	Lifecycle timestamps, nullable
tasks[].status_code	OPEN, APPROVED, SENT_BACK

12. Optional workflow operations

Recommended ownership

External source systems should normally create and reconcile requests. Human task approval, send-back and step completion should remain with Flowmate CA users unless a separately approved system-integration design authorizes automated processing.

API	Dummy payload	Response	Notes
POST /submitRequest	{"requestId":"<uuid>"}	Boolean value	Legacy explicit submission. Current UI does not call it.
POST /sendToS4	{"requestId":"<uuid>"}	accepted, integrationState, message	Currently a readiness boundary, not live PO transmission.
POST /addComment	{"requestId":"<uuid>","comment":"..."}	Comment entity	Add request comment.
POST /claimTeamTask	{"taskId":"<uuid>"}	Boolean value	Claim team task.
POST /approveTask	{"taskId":"<uuid>","remarks":"..."}	Boolean value	Approve an open task.
POST /sendBackTask	{"taskId":"<uuid>","targetStepNo":1,"remarks":"..."}	Boolean value	Send task back.
POST /completeStep	{"requestId":"<uuid>","stepNo":1,"remarks":"..."}	Boolean value	Complete step when mandatory tasks allow it.

sendToS4 dummy request and current response

```
{
  "requestId": "44444444-4444-4444-8444-444444444444"
}

{
  "accepted": false,
  "integrationState": "NOT_CONFIGURED",
  "message": "The S/4HANA integration boundary is ready. Configure the customer destination and API mapping before enabling transmission."
}
```

13. Error contract and troubleshooting

HTTP	Typical cause	Caller action
400	Invalid type/variant, invalid JSON details, invalid team, malformed OData input	Correct payload or master-data code; do not retry unchanged
401	Missing, expired or invalid bearer token	Obtain a new token and confirm target/audience
403	Missing authority or technical principal not provisioned in Users	Check granted CA scope and active shared user mapping
404	Request/task/resource not found	Verify UUID, environment and service root
409	Invalid lifecycle transition; e.g. submit non-draft request	Refresh resource and apply allowed transition
5xx	Service, HANA, destination or downstream failure	Retry only idempotent reads automatically; reconcile before retrying create

CAP/OData error example

```
{
  "error": {
    "code": "400",
    "message": "The selected request variant does not belong to this request type",
    "target": "input/requestVariantCode"
  }
}
```

Idempotency and retry guidance

The createRequest contract has no idempotency key and the generated reference number includes a random suffix. A network timeout after server commit can therefore produce an unknown outcome. Before retrying, query Requests using a source-system correlation value agreed with the Flowmate team (for example externalObjectId after a contract enhancement) or reconcile by title/requester/time window. Do not blindly repeat POST createRequest.

Validation gap

The backend currently validates request type, variant, processor team and JSON syntax, but does not uniformly enforce every PO field marked required by the UI. A technically successful HTTP response does not guarantee business completeness. Validate all Required* fields in the source system.

14. Implementation checklist

Area	Acceptance criterion
Security	Dedicated flowmate-api-auth service key stored in a vault; no secrets in source/logs
Authorization	Required CA authority granted to flowmate-api-client and accepted by Flowmate CA XSUAA
Provisioning	Technical JWT identity mapped to an active shared Users record
Connectivity	Direct service route or approved destination reachable from source system
Master data	Active value-help codes resolved and cached with refresh policy
Payload	details serialized exactly once as a JSON string
Validation	All Required* header and attachment fields validated before POST
Attachments	Metadata and binary upload retried/reconciled independently
Observability	Log correlation ID, request ID and referenceNumber; redact token and personal data
Retries	No blind retry of non-idempotent createRequest
Reconciliation	Read-back with purchaseOrder/items/attachments after creation
Testing	Negative tests for invalid code, expired token, unprovisioned user and failed file upload

Minimum test pack

Case	Expected result
Valid PO_OPEX with one item and four attachments	201/200 sequence; request and all files readable
Unknown requestVariantCode	400
Inactive processorTeamCode	400
Malformed details string	400
Unprovisioned technical user	403
Expired bearer token	401
Duplicate create after simulated timeout	Detected through reconciliation; no blind duplicate
Failed content PUT	Request remains; attachment retry process invoked

Appendix A - Swagger-to-runtime reconciliation

Area	Observed difference	Integration decision
NewRequestInput.processorTeamCode	Present in current CDS and UI; absent from checked-in Swagger	Required by integration flow
PO header fields after totalValue	Several current fields absent from Swagger create schema	Included in this guide from db/schema.cds
Additional PO item fields	Current model includes currency, contractNo, campaignLocationCode, siteId, plant, itemCategory, accountAssignment, materialGroup	Included in this guide
createBulkRequests	Present in current CDS/implementation but absent from current combined Swagger path set	Not recommended as primary PO integration contract until OpenAPI is refreshed
submitRequest behavior	Swagger exposes action; current UI creates directly and does not invoke it	Treat as legacy/optional

Appendix B - Source artifacts

Artifact	Purpose
flowmate/docs/flowmate-openapi.json	Combined OpenAPI 3.0.2 contract
flowmate-CA/srv/flowmate-ca.cds	Service entities, functions and actions
flowmate-CA/db/schema.cds	Persistent PO header/item data types
flowmate-CA/srv/flowmate-ca.js	Runtime validation and transaction behavior
flowmate-CA/app/flowmateca/webapp/model/FormDefinitions.js	UI mandatory fields, value helps and item columns
flowmate-CA/app/flowmateca/webapp/controller/RequestCreate.controller.js	Actual UI request and attachment sequence
flowmate-CA/xs-security.json	CA scopes and role templates
flowmate/xs-security-api.json	Technical API client XSUAA definition

Release note: Confirm the deployed service metadata, route, XSUAA grant and technical-user provisioning in the target environment before sharing credentials or starting end-to-end testing.